

ZAIN KHALIL

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EDUCATION

University of Waterloo

Bachelor of Computer Science (BCS), Co-op

Waterloo, ON

Fall 2025 – Present

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C, Racket, HTML/CSS

Backend & Data: FastAPI, REST APIs, React, Node.js/Express, PostgreSQL, MySQL, pgvector, Pydantic

Cloud & Infrastructure: AWS (EC2, S3, SES), Docker, Docker Compose, Linux, Bash, Nginx, GitHub Actions, Git, pytest

EXPERIENCE

Software Engineering Intern

May 2026 – Aug. 2026

IdealRatings

Cairo, Egypt

- Built and deployed a **Python** report-automation tool used by **~30 employees** to retrieve and process company reports, upload results to **AWS S3**, and generate output Excel files; automated **~95%** of sources across **~100-company** files, reducing workflows from **several days to a few hours**
- Engineered a modular **Python/MySQL** market-data pipeline targeting **~5,000 NYSE/Nasdaq securities**, integrating exchange and SEC data for prices, shares outstanding, fundamentals, news, and corporate actions with upserts, retries, failure isolation, and **85 automated tests**
- Developed and deployed a **FastAPI** email service using **Amazon SES**, implementing SHA-256 API-key authentication, MySQL audit logging, HTML/plain-text delivery, attachments, and AWS IAM/Boto3 integration
- Automated application deployment and operations using **Linux**, Bash, Nginx, systemd, Docker Compose, and GitHub Actions, adding backups, health checks, CI validation, SSH-based deployment, logging, and service verification

Executive Director - OMAM Model United Nations

Aug. 2024 – Mar. 2025

The International School of Elite Education

Cairo, Egypt

- Led **100+ members** and **40+ team heads** in delivering a conference for **~150 delegates** while managing a **~20,000 CAD** operating budget

PROJECTS

Abdo: Egyptian-Arabic Household Assistant | Python, FastAPI, PostgreSQL, pgvector, Railway Jun. 2026 – Present

- Built and deployed a Telegram-based household assistant supporting text, Egyptian-Arabic voice, live location, memory, calendar actions, shopping lists, and delivery tracking; handled **300+ interactions** in live use
- Designed an extensible **LLM tool-calling architecture** integrating Anthropic Claude, Cohere embeddings, Google Calendar, ElevenLabs STT/TTS, web search, and **pgvector RAG** without changing the core request loop for new capabilities
- Improved production reliability with asynchronous voice processing, two-layer webhook authentication, deterministic date resolution, embedding validation/backfills, and guarded external-service failures to prevent duplicate or fabricated actions

SpecGuard: AI Test Intelligence Platform | Python, FastAPI, React, TypeScript, OpenAI API

Mar. 2026

- Built a full-stack QA platform with a multi-stage **parse → extract → generate → validate → score** pipeline that converts product specifications into structured functional, edge-case, and negative test suites
- Enforced AI outputs using **Pydantic schemas** and automated retries, added heuristic coverage scoring across 4 quality metrics, and implemented **23 passing pytest validation tests**